Session 17: Generalised Linear Models **Practical**

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Practical overview

Understand and predict the number of runners at “parkrun” events

1. Data exploration
2. Univariable Poisson models
3. Checking assumptions – is a Poisson model adapted?
4. Multivariable model
5. Making predictions

Data exploration

Most already given in practical question.

Note that the trend is not easy to see.

There may be interactions between location and other variables such as humidity and temperature.

Univariable Poisson models

Temperature, location, year (2016) and month (if tested) are all significant variables.
Relative humidity has p = 0.963 so we will not include it in a multi-variable model.

You should have outcomes such as the one below:

```
Call:
glm(formula = total ~ tempc, family = poisson, data = dat)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.5177039  0.0046826 964.779  < 2e-16 ***
tempc       -0.0042604  0.0008058  -5.287 1.24e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1897.7  on 537  degrees of freedom
AIC: 5313.9

Number of Fisher Scoring iterations: 4
```

Checking assumptions

Assumption: the total number of runners follows a Poisson distribution.
Let us develop a model with just the intercept:

```
Call:
glm(formula = total ~ 1, family = poisson, data = dat)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 4.510860   0.004515     999   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1925.6  on 538  degrees of freedom
AIC: 5339.9

Number of Fisher Scoring iterations: 4
```

```
Call:
glm(formula = total ~ tempc, family = poisson, data = dat)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.5177039  0.0046826 964.779  < 2e-16 ***
tempc       -0.0042604  0.0008058  -5.287 1.24e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1897.7  on 537  degrees of freedom
AIC: 5313.9

Number of Fisher Scoring iterations: 4
```

Note: the deviance of this model is the same as the null deviance in this and other models, which is logical (hence, there was no need to develop the null model).

Let us undertake a chi-squared test on the deviance of the null model:
`pchisq(1925.6,538,lower.tail=FALSE)` `4.245051e-155`

There is considerable evidence against the null (this would not change substantially if we used any of the univariable models we have developed).

Multivariable model

A quasipoisson is best adapted (there is considerable evidence against the null, and when using quasipoisson we see that the dispersion parameter is between 3 and 4.

Note: the AIC is NA because quasipoisson models do not have fully specified likelihoods.

```
Call:
glm(formula = total ~ rhc + tempc + loc + year + month, family = quasipoisson,
    data = dat)

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.6838427  0.0332074 141.048  < 2e-16 ***
rhc              -0.0004716  0.0009396  -0.502  0.61597
tempc            -0.0019581  0.0028186  -0.695  0.48754
loccrystalpalace -0.0367790  0.0236959  -1.552  0.12124
locfinsbury      -0.0594888  0.0233012  -2.553  0.01096 *
lockingston      -0.0361426  0.0226449  -1.596  0.11108
year2016         -0.0405638  0.0195483  -2.075  0.03847 *
year2017         -0.0334070  0.0210847  -1.584  0.11371
month02          -0.0877310  0.0356644  -2.460  0.01422 *
month03          -0.1092387  0.0387827  -2.817  0.00504 **
month04          -0.0918128  0.0387081  -2.372  0.01806 *
month05          -0.0579681  0.0453201  -1.279  0.20144
month06          -0.0784568  0.0509016  -1.541  0.12384
month07          -0.1688860  0.0543704  -3.106  0.00200 **
month08          -0.1786112  0.0536348  -3.330  0.00093 ***
month09          -0.0877600  0.0451232  -1.945  0.05233 .
month10          -0.0974511  0.0419540  -2.323  0.02057 *
month11          -0.1756292  0.0426857  -4.114 4.51e-05 ***
month12          -0.2985858  0.0439660  -6.791 3.05e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for quasipoisson family taken to be 3.190915)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1632.8  on 520  degrees of freedom
AIC: NA
```

Multivariable model

You may apply robust standard errors on quasi-poisson:

```
library(sandwich)
cov <- vcovHC(m1,type="HC0")
coeftest(m1, vcov. = vcovHC(m1, type = 'HC0'))
se <- sqrt(diag(cov))
robust.SE <- cbind(Estimate= coef(m1), "Robust SE" = se, "Pr(>|z|)" = 2 *
pnorm(abs(coef(m1)/se),lower.tail=FALSE))
```

	Estimate	Robust SE	Pr(>|z|)
(Intercept)	4.6838426700	0.042529090	0.000000e+00
rhc	-0.0004715539	0.001053404	6.544073e-01
tempc	-0.0019581413	0.002918979	5.023283e-01
loccrystalpalace	-0.0367790420	0.021907253	9.318083e-02
locfinsbury	-0.0594888198	0.021758876	6.256959e-03
lockingston	-0.0361425989	0.023478543	1.237095e-01
year2016	-0.0405638392	0.020101726	4.359874e-02
year2017	-0.0334070125	0.021518331	1.205448e-01
month02	-0.0877309801	0.035424745	1.326608e-02
month03	-0.1092386753	0.039584154	5.786213e-03
month04	-0.0918128120	0.044761433	4.025147e-02
month05	-0.0579680954	0.049498756	2.415578e-01
month06	-0.0784567811	0.070018687	2.624956e-01
month07	-0.1688859866	0.061574720	6.092126e-03
month08	-0.1786111677	0.059193333	2.549360e-03
month09	-0.0877599636	0.046250385	5.776238e-02
month10	-0.0974510517	0.043258703	2.427490e-02
month11	-0.1756291615	0.048081175	2.594332e-04
month12	-0.2985858489	0.048516646	7.541127e-10

Making predictions

There are going to be considerable errors around predicted values.

Create a dataset with:

- Each location
- Year = 2017
- Month = 05 (May)
- Temperature = 10 degrees so centered temperature = 0
- Relative humidity = 70 so centered relative humidity = 0

Predict the expected value for the number of runners under the specified conditions, and the confidence interval for that expected value:

```
newdat <- data.frame(loc=c("brockwell","crystalpalace","finsbury","kingston"),
tempc=0, rhc=0, year=as.factor(2017), month=as.factor("05"))
tmp <- predict(m1, newdat, se.fit=TRUE)
newdat$pred <- exp(tmp$fit)
newdat$low <- exp(tmp$fit -1.96 * tmp$se.fit)
newdat$high <- exp(tmp$fit + 1.96 * tmp$se.fit)
```

Making predictions

The organizers probably want the uncertainty around the number of runners that will come on a specific occasion i.e. the prediction interval. It can best be obtained through simulation:

```
newdat$pred.avg <- 0
newdat$pred.lwr <- 0
newdat$pred.upr <- 0
for(i in 1:dim(newdat)[1]){
sim <- rpois(1000,rnorm(1000,newdat$pred[i],tmp$se.fit[i]))
newdat$pred.avg[i] <- round(quantile(sim,0.5),2)
newdat$pred.lwr[i] <- round(quantile(sim,0.025),2)
newdat$pred.upr[i] <- round(quantile(sim,0.975),2)
}
```

loc <chr>	tempc <dbl>	rhc <dbl>	year <fctr>	month <fctr>	pred <dbl>	low <dbl>	high <dbl>	pred.avg <dbl>	pred.lwr <dbl>	pred.upr <dbl>
brockwell	0	0	2017	05	98.73777	92.15428	105.7916	99	80	119.02
crystalpalace	0	0	2017	05	95.17226	89.01751	101.7526	95	77	115.00
finsbury	0	0	2017	05	93.03528	86.82548	99.6892	93	75	113.00
kingston	0	0	2017	05	95.23285	88.99027	101.9133	95	77	115.02